เอกสารข้อกำหนดระบบ (SYSTEM REQUIREMENTS)

System Requirements Specification

Phase 1 — SA output — system-level requirements extending REQ (IEEE 830-style)

DOCUMENT CODE	DFC-SRS-001
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PROJECT	Daily Fitness Companion (Phase 1)

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DOCUMENT CONTROL

ข้อมูลควบคุมเอกสาร

Control record for System Requirements Specification

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02 APPROVAL & OWNERSHIP *การอนุมัติและความเป็นเจ้าของ*

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03 LIFECYCLE *วงจรชีวิตเอกสาร*

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04 REVISION HISTORY *ประวัติการแก้ไข*

DATE / วันที่	REV	DESCRIPTION / รายละเอียดการแก้ไข	BY / โดย
27 July 2026	0.7	v0.7: แก้ consistency ทั้งชุด — SRS solo dev (เดิม 2 คน), REQ SCS→SCH + ปรับเลข WBS ให้ตรงรหัสจริง + เติม traceability §5 ครบ (เพิ่ม F-014/015, N-003/004/006/007/008/010, R-005) + SA Reference จาก SRS, PRD dead refs (sprint-plan/tasks)→WBS/EST/SCH/HND/SRS, HND version→v0.7 + แก้เลข section, เพิ่ม US-14/US-15 เข้า WBS+EST (base 80.5→86.5) propagate SCH, reconcile effort, resume window 7→30 วัน, analytics event naming, cohort size, SRS §4.4 priority, onboarding 12+medical	Product Owner
25 July 2026	0.6	เข้า SA phase — เพิ่ม SRS v0.1 (System Requirements Specification, IEEE 830-style) จาก REQ + HND. Q1/Q2 marked as blocking. Next: SAD after Q1/Q2 decided	Product Owner
25 July 2026	0.5	เพิ่ม HND (Handover Document) เตรียมส่งมอบ SA + เพิ่ม §6 System Context (External Interfaces / User Classes / Operating Environment / Data Volume / Quality Attribute Priorities) ให้ SA มี input พอสำหรับ SRS+SAD ตาม IEEE 830. อัปเดต REQ traceability เพิ่ม SA Reference column	Product Owner
25 July 2026	0.4	จัดใหม่ตาม workflow PRD→REQ→WBS→EST→SCH. เพิ่ม 4 เอกสาร standard PM: REQ (requirements catalog + traceability), WBS (hierarchical work breakdown, 100% rule), EST (effort estimates with confidence levels), SCH (schedule framework template, owner-fills). Archive sprint-plan และ sprint-tasks (redundant กับ WBS+EST+SCH)	Product Owner
25 July 2026	0.3	ถอด resource assumptions (13 สัปดาห์, 8-10 ชม./สัปดาห์, F&F 5 คน, buffer %, sprint dates) ออกจาก sprint-plan และ sprint-tasks. เปลี่ยนเป็น Work Packages (WP0-WP6) ที่ไม่ผูก schedule. เก็บ effort estimates ไว้เป็น work sizing. Owner จัดการ scheduling เอง.	Product Owner
25 July 2026	0.2	ปรับ layout หน้า 1 (Cover) ใช้ CSS Grid 5 rows + หน้า 2 (Doc Control) ปรับ section markers เป็น badge สีดำ, compress spacing	Product Owner
25 July 2026	0.1	ฉบับแรก (Initial release) — Cover page, doc control, body content	Product Owner

System Requirements Specification (SRS) — Daily Fitness Companion (Phase 1)

Owner: Systems Analyst (SA) · **Status:** Draft v0.1 · **Last updated:** 2026-07-25 **Position in workflow:** PRD → REQ → WBS → EST → SCH → HND → **SRS** → SAD → DDD/API/DB/Sec/Deploy **Source docs:** PRD, REQ (traceability), HND (system context) **Consumed by:** SAD (architecture), Dev team

> ***Note:*** *SRS นี้ extend REQ ด้วย system-level detail. Tech-specific decisions (platform, backend, framework) ที่รอ Q1/Q2 answer จะ mark เป็น* ***TBD*** *และ SA จะเติมหลัง ADR ใน SAD*

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1. Introduction

1.1 Purpose

เอกสารนี้ระบุ system-level requirements ที่ **Development team** จะใช้ build Daily Fitness Companion Phase 1 — ครอบคลุมทั้ง functional, external interfaces, data model, non-functional, และ constraints ในระดับที่ชัดเจนพอให้ implement ได้ ตรงตาม spec โดยไม่ต้องตีความเพิ่ม (solo developer)

1.2 Scope

In-scope (Phase 1): - User-facing mobile app (Today view + edit + browse) - Owner-facing admin (submission queue + plan input) - Backend persistence + notifications - 5-domain plan structure (workout / meal / supplement / cardio / recovery)

Out-of-scope (Phase 2+): - LLM auto-generation (SA design ต้องเผื่อรองรับ) - User authentication (Phase 1 = device-only) - Payment / subscription (Phase 2) - Advanced features per PRD Non-goals

1.3 Definitions

Term	Meaning
User (Nan)	End user — fitness member ที่รับ plan + ใช้ Today view
Owner	Wizard-of-Oz coach + system admin (Phase 1: 1 person)
Submission	Onboarding form ที่ user ส่งเข้าระบบ
Plan	Weekly schedule ครบ 5 domain ที่ owner generate ให้ user
Day	ข้อมูลต่อวันใน plan (workout/meal/supp/cardio/recovery ของวันนั้น)
Item	หน่วยย่อยใน Day (เช่น 1 exercise, 1 meal, 1 supplement)
PDPA	Personal Data Protection Act B.E. 2562 (Thailand privacy law)
NFR	Non-Functional Requirement
AC	Acceptance Criterion

1.4 References

Doc	Purpose
PRD.md	Product vision + business goals
REQ.md	Requirements catalog (47 items, traced from PRD)
HND.md	Handover package + System Context §6
Stories.md	User stories with Given/When/Then AC
WBS.md / EST.md / SCH.md	Implementation planning

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2. Overall Description

2.1 Product Perspective

Type: Greenfield mobile-first application, standalone (no legacy system)

Deployment: Client (mobile app) + Backend API + Managed backend service

Approach for Phase 1: Wizard-of-Oz — Owner uses external ChatGPT UI to generate plans; input via admin queue. No LLM API integration ใน Phase 1

2.2 Product Functions (Extends REQ §2)

User-facing: 1. Onboarding submission (12-Q form + save/resume + medical block) 2. Today view (5-domain glanceable view + browse days + offline) 3. Item editing (workout/meal/supp modal edits) 4. Duplicate week (stretch)

Owner-facing: 1. Submission queue (list + sort + filter) 2. Plan input (5-domain structured input + auto-save) 3. Mark plan ready + notify user

Cross-cutting: 1. Disclaimer at key moments 2. Analytics event logging (non-PII) 3. Email notifications (owner + user)

2.3 User Classes and Characteristics (Extends HND §6.2)

Class	ID	Access level	Frequency	Technical background
Fitness User	U1	Standard (own data only)	Daily (2–4 opens/day)	Tech-literate, smartphone daily
Owner / Admin	U2	Elevated (all data + queue)	Weekly (per plan generation)	Owner = SA/dev = high tech literacy
Anonymous Visitor	U3	Read landing/onboarding only	One-time	Varies

Note: U1 access to own data enforced by device-only storage (Phase 1) → Phase 2 auth + row-level security

2.4 Operating Environment (Extends HND §6.3)

Client environment: - **Target platform:** TBD — pending Q1 decision (iOS native / PWA / RN / Flutter) - **Mobile OS minimum:** iOS 15+ / Android 10+ - **Screen sizes:** responsive 320px – 1920px - **Network:** offline-first for Today view; online required for submit + owner input - **Peak load:** morning 7–9am, before workout 5–7pm, before meals

Server environment: - **Backend platform:** TBD — pending Q2 decision (Firebase / Supabase / custom) - **Region:** Southeast Asia (Singapore recommended) - **Availability target:** ≥ 99% uptime (REQ-N-010)

Development environment: - OS: macOS (owner's dev machine) - CI/CD: platform-managed (GitHub Actions or hosting built-in)

2.5 Constraints (Consolidated from HND §5)

Technical: - Solo developer — no coordination overhead but no parallelization - Must be buildable + maintainable by 1 person - Offline-first for Today view (REQ-N-002) - Health data local storage priority (REQ-N-004) - ≤ 500ms Today view render (REQ-N-001)

Business: - Personal project — budget = free tier where possible - Timeline: TBD (owner-managed scheduling)

Regulatory: - PDPA compliance (REQ-R-005) - Medical disclaimer (REQ-R-001) - Medical condition block/bypass (REQ-R-002) - WCAG 2.1 AA basics (REQ-N-005) - Thai + English locale (REQ-N-006)

2.6 Assumptions and Dependencies

#	Assumption	Impact if false
A1	Tech stack familiar to owner	Effort estimate +25%
A2	Backend-as-a-service ตอบโจทย์	Custom backend = large effort
A3	Email notification เพียงพอ (no push)	UX degradation
A4	F&F cohort เข้าใจ Wizard-of-Oz	Positioning gap
A5	Phase 1 ไม่ต้อง user account	Migration pain ตอน Phase 2
A6	Owner has ChatGPT access	Alternative LLM required

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3. External Interface Requirements

3.1 User Interfaces

Design system: TBD — SA + PO to decide (custom / native OS defaults / library)

Key screens (extends Stories.md):

#	Screen	User class	Reference
3.1.1	Onboarding intro + disclaimer	U1, U3	US-01, US-16
3.1.2	12-question form (paginated)	U1	US-01, US-02
3.1.3	Medical block screen	U1	US-03
3.1.4	Pending “waiting for owner” state	U1	US-11
3.1.5	Today view (main app screen)	U1	US-08, US-09, US-10
3.1.6	Week navigator + browse	U1	US-12
3.1.7	Edit item modal (workout/meal/supp/cardio)	U1	US-13, US-14
3.1.8	Empty + rest day states	U1	US-11
3.1.9	About / Terms / Privacy	U1	US-16
3.1.10	Admin — submission queue	U2	US-04
3.1.11	Admin — plan input (5 domain)	U2	US-05, US-06
3.1.12	Admin — mark ready action	U2	US-07

3.2 Hardware Interfaces

- **Touch screen:** required (mobile-first)
- **Network:** Wi-Fi / cellular
- **Storage:** local device storage for offline cache (~10 MB typical)
- **No hardware sensor integration** (no wearables in Phase 1 — see PRD non-goals)

3.3 Software Interfaces (Extends HND §6.1)

#	External System	Direction	Protocol	Auth	Failure handling	Phase
E1	**Owner email inbox**	Outbound	SMTP / provider API (TBD: SendGrid / Resend / Postmark)	API key	Retry 3× with exponential backoff; alert owner via secondary channel if fail	1
E2	**User email inbox**	Outbound	SMTP / provider API	API key	Retry 3×; log failure; user can request resend via admin	1
E3	**ChatGPT (owner browser)**	Manual UI	Web app	Owner's account	Owner action, no system dep	1
E4	**LLM API**	Outbound (Phase 2 only)	HTTPS REST/streaming	API key + rate limit	Retry with backoff; fallback to cached prompt result if available	2
E5	**Payment gateway** (Phase 2)	Outbound + webhook	HTTPS + HMAC-signed webhook	API key + webhook secret	Provider handles idempotency	2
E6	**Analytics platform** (Phase 1+2)	Outbound events	HTTPS REST	API key	Fire-and-forget with local buffer; drop on repeated fail	1+2
E7	**Backend service** (Phase 1+2)	Bi-directional	Provider SDK / HTTPS REST	(see auth spec Phase 2)	Retry idempotent operations; user-visible error otherwise	1+2

#	External System	Direction	Protocol	Auth	Failure handling	Phase
E8	**Push notification** (optional)	Outbound	FCM / APNs	Provider credentials	Fire-and-forget; log undeliverable	1+

Detailed contracts: SA จะเขียนใน API Spec doc (ต่อไป)

3.4 Communications Interfaces

- **All external traffic HTTPS** (TLS 1.2+)
- **HSTS** for public web (if PWA chosen)
- **Certificate management** ผ่าน hosting provider (Let's Encrypt / managed)
- **API auth (Phase 2):** Bearer JWT ใน `Authorization` header
- **Webhook auth:** HMAC-SHA256 body signature

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4. System Features

Features grouped by workflow. Each block extends REQ-F with system-level detail.

4.1 Onboarding & Submission (WP 1.2)

Traces to: REQ-F-001, REQ-F-002, REQ-F-003, REQ-B-007 **Priority:** P0

4.1.1 Description

ระบบต้องรับ user profile ผ่าน 12-question form, validate, persist, notify owner, และแสดง "pending" state จนกว่า owner จะ mark plan ready

4.1.2 Preconditions

- User เปิดแอปครั้งแรก
- Network available (สำหรับ submit — form filling OK offline)

4.1.3 Main Flow

1. แสดง welcome screen + disclaimer (REQ-R-001) — user ยอมรับ
2. Present 12-question form paginated (1 Q per screen)
3. On each question: validate input inline; on invalid, block "next" button
4. On step change: save partial state to local storage (REQ-F-002)
5. On question 12 → present review + submit button
6. On submit: POST to backend endpoint

7. Backend persists submission + triggers email to owner (E1)
8. Show “pending — รอ owner ~24 ชม.” screen (until status changes)

4.1.4 Alternate Flows

AF1: Medical condition selected (REQ-F-003) - User selects Yes on any medical Q (heart/diabetes/injury/pregnancy) - System shows block screen + guidance “ปรึกษาแพทย์” - User can: (a) Cancel submission, or (b) Explicit bypass with checkbox + confirmation modal - If bypass: submit proceeds with `medical_bypass_consent = true` - Owner queue shows flag

AF2: Network fail on submit - Retry 3× with exponential backoff - On persistent fail: show error with “ลองใหม่” button; form state preserved - No submission recorded until successful ACK from backend

AF3: Resume partial (REQ-F-002) - User closes app mid-form - Reopen within 30 days: form restores at last completed step - Reopen after 30 days: partial state discarded, restart from step 1

4.1.5 Data Touched

- **Write:** `submissions` (new record with status=‘pending’)
- **Read/write:** local storage `onboarding_partial` (temp until 30d)
- **Trigger:** email notification (E1) to owner

4.1.6 Verification

- Integration tests for form validation
- E2E test for full submit → owner sees in queue
- Manual test for medical block + bypass
- Timing test for partial save (should be instant, ≤ 100ms)

4.2 Owner Submission Queue (WP 1.2)

Traces to: REQ-F-004 **Priority:** P0

4.2.1 Description

Owner ต้องเข้าถึง admin view เพื่อดู submissions ทั้งหมด, sort ตาม pending oldest first, click เข้าไปดูรายละเอียดของแต่ละคน

4.2.2 Preconditions

- Owner accesses admin URL (Phase 1: password-protected via .env secret)
- Backend accessible

4.2.3 Main Flow

1. Owner navigates to `/admin/queue`
2. Password prompt (single value from env; Phase 2 → proper auth)
3. On success: fetch submissions list from backend
4. Render list: name, email, submitted_at, status, medical_flag
5. Sort: `pending` first, oldest submitted_at first
6. Owner clicks row → detail view (§4.3 Plan Input)

4.2.4 Alternate Flows

AF1: Empty queue - Show “ยังไม่มี user ใหม่” empty state + link to test-generate

AF2: Auth fail - Rate-limit password attempts (5 fails / hour) - Log failed attempts

4.2.5 Verification

- Unit test for sort order
- Integration test for auth
- Manual test for empty state

4.3 Owner Plan Input — 5 Domain (WP 1.3)

Traces to: REQ-F-005, REQ-F-006, REQ-F-007 **Priority:** P0

4.3.1 Description

Owner กรอก plan ครบ 5 domain (workout / meal / supplement / cardio / recovery) สำหรับ submission ที่เลือก, auto-save ระหว่าง input, จากนั้นกด “Mark ready” เพื่อ notify user

4.3.2 Preconditions

- Owner ผ่าน auth (§4.2)
- Selected submission มี status=‘pending’

4.3.3 Main Flow

1. Show plan input UI แบ่งเป็น 5 tabs หรือ sections
2. **Workout tab:** day picker Mon–Sun + rest toggle + exercises (name/sets/reps/rest/note) + reorder
3. **Meal tab:** per day → meals (mealtime/food items/portion/optional macros)
4. **Supplement tab:** items (name/dose/timing) applied to relevant days
5. **Cardio tab:** type/duration/frequency

6. **Recovery tab:** sleep target + rest day count + mobility notes
7. On every field change: debounced auto-save (500ms)
8. Once complete: owner clicks “Mark ready”
9. System validates workout section not empty
10. Status → `plan_ready`, timestamp saved
11. Trigger email to user (E2)
12. User’s next app-open routes to Today view

4.3.4 Alternate Flows

AF1: Auto-save fail (network) - Show inline warning “ยังไม่ได้ save — retry…” - Buffer changes locally, retry on reconnect - Do NOT lose owner input

AF2: Mark ready with missing workout - Block action, show error “ยังต้องกรอก workout อย่างน้อย 1 วัน”

AF3: Owner reopens partially-filled plan - Load latest saved state - All previously entered data intact

4.3.5 Data Touched

- **Write:** `plans`, `plan_days`, `plan_items` (per domain)
- **Update:** `submissions.status`
- **Trigger:** email E2

4.3.6 Verification

- Integration tests for auto-save (debounce + retry)
- E2E for full plan input → user receives email
- Manual for reorder + validation

4.4 Today View (WP 1.4 + 1.5)

Traces to: REQ-F-008, REQ-F-009, REQ-F-010, REQ-F-011, REQ-B-008 **Priority:** P0 (F-008, F-009) / P1 (F-010, F-011) ⚠️ **Risk R2:** biggest task, UX polish blow-up risk

4.4.1 Description

User เปิดแอปเห็น plan ของวันนี้ทันที (< 5 วิ perceived) ครบ 5 domain เรียงตามเวลา, glanceable, offline-capable

4.4.2 Preconditions

- User มี plan status=‘plan_ready’ หรือ ‘delivered’

- Local cache populated (or online fetch OK)

4.4.3 Main Flow

1. On app-open: display skeleton loader (< 100ms)
2. Read today's day from local cache (offline-first)
3. Determine day type: workout / rest / meal-only / no plan
4. Render sections in time-of-day order:
    - Morning meal (if any)
    - Pre-workout supplement (if any)
    - Workout section (if workout day)
    - Post-workout meal + supplement
    - Additional meals + supplements
    - Cardio (if any)
    - Recovery / sleep target (bottom)
5. Fire analytics event `today_view_opened` with `time_of_day`
6. Background sync with backend (if online) to refresh cache

4.4.4 Alternate Flows

AF1: Rest day - Show "Rest day — พักผ่อน" prominent - Meals still displayed if defined

AF2: Offline - Serve from cache - Show subtle offline indicator (top badge) - No error UX

AF3: Plan not ready (pending) - Show pending state (§4.1.3 step 8) with email support link

AF4: No plan (never onboarded) - Show CTA "Start onboarding"

AF5: Cache miss + offline - Show generic "no internet, tap to retry" screen - Should be extremely rare — cache populated on first successful load

4.4.5 Data Touched

- **Read:** local cache (or backend if cache miss)
- **Write:** analytics events (async)
- **Update:** local cache from background sync

4.4.6 Verification

- **REQ-N-001:** performance test — render ≤ 500ms p95 on mid-tier phone
- **REQ-N-002:** airplane mode test — full functionality

- E2E for all state variations (workout / rest / pending / no plan)
- Manual UX review — “glanceable” criterion

4.5 Browse Days + Edit Items (WP 1.5)

Traces to: REQ-F-012, REQ-F-013, REQ-F-014, REQ-F-015 **Priority:** P0 (US-12, US-13) / P1 (US-14) / P2 (US-15)

4.5.1 Description

User navigate ระหว่างวันของสัปดาห์ + แก้ / เพิ่ม / ลบ item ในแต่ละวัน

4.5.2 Main Flow

Browse (REQ-F-012): 1. Week navigator แสดง Mon–Sun (7 pills or swipe) 2. Tap/swipe → เปลี่ยน selected day 3. Fetch that day from cache 4. Persist selection (last viewed day)

Edit item (REQ-F-013): 1. Long-press or tap item → context menu (Edit / Delete) 2. Edit → modal with editable fields (varies by item type) 3. Save → optimistic update + backend sync 4. Cancel → discard (with unsaved warning if dirty)

Add item (REQ-F-014): 1. In section footer: “+ เพิ่ม item” 2. Same modal as Edit but empty 3. Save → append to section

Duplicate week (REQ-F-015, P2): 1. Menu action: “Duplicate this week” 2. Confirm modal (“จะเขียนทับ plan สัปดาห์หน้า?” if applicable) 3. Copy operation (backend transaction)

4.5.3 Data Touched

- **Read:** cache (day-specific)
- **Write:** cache + backend on save (optimistic)

4.5.4 Verification

- E2E: edit workout, verify persists across reload
- Unit tests for cache mutation logic
- Manual: unsaved-changes warning

4.6 Safety & Disclaimer (Cross-cutting)

Traces to: REQ-R-001, REQ-R-002, REQ-F-016 **Priority:** P0 (safety-critical)

4.6.1 Description

Disclaimer “ไม่ใช่คำแนะนำทางการแพทย์” ต้องปรากฏที่จุดสำคัญ + medical condition ต้องถูก handle เข้ม

4.6.2 Points of enforcement

1. **Onboarding intro screen** — full disclaimer + “ยอมรับและเริ่ม” button
2. **First plan review** — banner disclaimer (dismissible, saved state)
3. **About page** — link to full Terms + Privacy + Disclaimer
4. **Medical condition question** — if flagged → block screen (§4.1.4 AF1)

4.6.3 Copy source

Reference to `constants/disclaimer.ts` (single source of truth for Thai + English)

4.6.4 Verification

- Manual verification of copy at each point
- E2E: onboarding cannot proceed without accepting disclaimer
- Legal review before public launch (Phase 2)

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5. Non-Functional Requirements (Extends REQ §3)

5.1 Performance

#	Requirement	Target	Measurement method	Trace
N-1.1	Today view render	p95 ≤ 500ms (cached, mid-tier phone)	Automated perf test (Lighthouse หรือ equivalent)	REQ-N-001
N-1.2	Onboarding form navigation	≤ 200ms per step	Manual + integration timing	REQ-N-003
N-1.3	Local storage read/write	≤ 50ms	Unit test	Implicit
N-1.4	Backend API response (typical)	p95 ≤ 500ms	APM (Application Performance Monitoring)	Implicit
N-1.5	Cache warm-up (post-login)	≤ 2s	Manual timing	Implicit

5.2 Availability

#	Requirement	Target	Measurement	Trace
N-2.1	Backend uptime (Phase 1)	≥ 99% monthly	Uptime monitor	REQ-N-010
N-2.2	Today view offline capability	100% functionality (cached)	Airplane mode test	REQ-N-002
N-2.3	Recovery time (backend down)	Full recovery ≤ 30 min	Incident response	Implicit

5.3 Scalability (Phase 1 targets — SA design for Phase 2)

#	Metric	Phase 1	Phase 2 target	Trace
N-3.1	Concurrent users	≤ 10	10 – 50 peak	HND §6.4
N-3.2	Total registered users	≤ 10	100 – 1,000	HND §6.4
N-3.3	Requests / user / day	≤ 10	5 – 10	HND §6.4
N-3.4	DB size	< 10 MB	< 10 GB (year 1)	HND §6.4

5.4 Security

Reference: **Security Architecture Doc** (ต่อไป). Key SRS-level requirements:

#	Requirement	Trace
N-4.1	All external traffic HTTPS (TLS 1.2+)	Implicit / industry standard
N-4.2	Health data local storage priority; backend encrypted at rest	REQ-N-004
N-4.3	Consent flow แจ้งชัดเรื่อง owner viewing (Phase 1)	REQ-N-007
N-4.4	Analytics events ไม่ track PII (email, name, health data)	REQ-N-008
N-4.5	Admin auth via .env secret Phase 1; proper auth Phase 2	Implicit

5.5 Maintainability

#	Requirement	Verification
N-5.1	Code test coverage ≥ 70% for business logic	Coverage report
N-5.2	API documented per API Spec doc	Doc review
N-5.3	ADRs (Architecture Decision Records) for major decisions	Present in SAD
N-5.4	Migrations idempotent + reversible	Migration review

5.6 Usability

#	Requirement	Target	Trace
N-6.1	WCAG 2.1 AA basics (contrast, touch ≥ 44px, keyboard, screen reader)	Automated (axe) + manual	REQ-N-005
N-6.2	Locales: Thai + English	2 locales supported	REQ-N-006
N-6.3	Time to answer “วันนี้ทำอะไร”	≤ 5 วิ (open → info visible)	REQ-B-002

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6. Data Requirements

6.1 Logical Data Model (details ใน DDD)

Core entities:

```
User ─┬─→ Submission ─→ Plan ─→ Day ─→ Item
      └─→ Preferences (Phase 2)

Item polymorphic per domain:
- WorkoutItem (exercise + sets + reps + rest + note)
- MealItem (mealtime + food + portion + macros)
- SupplementItem (name + dose + timing)
- CardioItem (type + duration + frequency)
- RecoveryItem (sleep target + mobility)
```

6.2 Data Volume (Extends HND §6.4)

See §5.3 Scalability targets.

6.3 Data Retention & Deletion

Entity	Active retention	Deletion trigger	Hard-delete SLA
User + submissions + plans	Indefinite (Phase 2: until sub cancelled + 30d)	User request or account close	Within 30 days (PDPA)
Local storage cache	30 days for onboarding partial; indefinite for plan cache	Explicit clear or 30d unused	Immediate on trigger
Analytics events	Aggregate: 12 months; raw: 30 days	Retention window expiry	Automatic (provider default)
Email logs	90 days	Retention window	Automatic

Compliance: REQ-R-005 (PDPA), REQ-R-004 (T&C before public)

6.4 Data Volume Growth Assumptions

- Per user: ~1 MB storage growth
- Analytics events: ~10–30 / user / day (aggregate stored)
- Backup: daily automated (backend provider default)

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7. Other Requirements

7.1 Regulatory

- **PDPA compliance** (Thailand) — REQ-R-005
- **Medical disclaimer** at all UI touchpoints — REQ-R-001
- **PCI DSS** (Phase 2, via payment gateway) — REQ-R-006

7.2 Localization / i18n

- Primary locale: Thai (`th-TH`)
- Secondary: English (`en-US`)
- Copy externalized in `i18n/{locale}.json`
- Number/date formatting per locale

7.3 Accessibility

- WCAG 2.1 AA basics per REQ-N-005
- Screen reader labels on all interactive elements
- Focus management for modals + forms
- No color-only information encoding (use icon + text)

7.4 Analytics Events (Phase 1 event catalog)

Event name	Trigger	Properties (non-PII)
`app_open`	Any app open	`time_of_day`, `day_of_week`
`onboarding_started`	User starts form	(none)
`onboarding_step_completed`	Each Q done	`step_number`
`onboarding_submitted`	Form submit success	`q_count`, `medical_flag`, `bypass`
`plan_delivered`	Owner marks ready	(server-side)
`today_view_opened`	Today view load	`time_of_day`, `day_type`
`item_edited`	User edits item	`item_type`
`item_added`	User adds item	`item_type`
`item_removed`	User removes item	`item_type`
`disclaimer_accepted`	User accepts	(none)
`rest_timer_used`	Should tier	(none, Phase 1.1+)

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8. Traceability Matrix

REQ → SRS Section → Test Type

REQ ID	Description	SRS Section	Test Type
REQ-B-001	Discovery pain solved	§1.2 scope	User interview
REQ-B-002	Reminder pain solved	§5.6 usability (N-6.3)	Manual timing
REQ-B-003	Wizard-of-Oz validation	§2.1 approach	Retro post-Phase 1
REQ-B-007	Onboarding completion ≥ 70%	§4.1	Funnel analytics
REQ-B-008	Adherence ≥ 80%	§4.4	Event analytics
REQ-F-001	Onboarding form	§4.1	Integration + E2E
REQ-F-002	Save/resume	§4.1.4 AF3	Integration
REQ-F-003	Medical block	§4.1.4 AF1, §4.6	Manual + E2E
REQ-F-004	Owner queue	§4.2	Integration + Manual
REQ-F-005	Owner workout input	§4.3	E2E
REQ-F-006	Owner nutrition input	§4.3	E2E
REQ-F-007	Mark ready + notify	§4.3	E2E
REQ-F-008	Today view workout	§4.4	E2E + performance
REQ-F-009	Today view meals	§4.4	E2E
REQ-F-010	Supp/cardio/recovery view	§4.4	E2E
REQ-F-011	Empty/rest states	§4.4 AF	E2E
REQ-F-012	Browse days	§4.5	E2E
REQ-F-013	Edit item	§4.5	E2E
REQ-F-014	Add/remove	§4.5	E2E
REQ-F-015	Duplicate week	§4.5	Manual
REQ-F-016	Disclaimer	§4.6	Manual + E2E
REQ-N-001	Today view ≤ 500ms	§5.1 N-1.1	Perf test
REQ-N-002	Offline-first	§5.2 N-2.2, §4.4 AF2	Airplane mode

REQ ID	Description	SRS Section	Test Type
REQ-N-003	Onboarding nav ≤ 200ms	§5.1 N-1.2	Timing
REQ-N-004	Health data local	§5.4 N-4.2, §6.3	Code review
REQ-N-005	WCAG 2.1 AA	§5.6 N-6.1, §7.3	Axe + manual
REQ-N-006	Thai + English	§5.6 N-6.2, §7.2	Locale switch
REQ-N-007	Owner viewing consent	§5.4 N-4.3	Consent screen present
REQ-N-008	No-PII analytics	§5.4 N-4.4, §7.4	Analytics review
REQ-N-010	Backend ≥ 99% uptime	§5.2 N-2.1	Uptime monitor
REQ-R-001	Medical disclaimer	§4.6, §7.1	Manual UI review
REQ-R-002	Medical block/bypass	§4.1.4 AF1, §4.6	E2E + manual
REQ-R-005	PDPA compliance	§6.3, §7.1	Legal review

Phase 2 REQs (REQ-F-017 to 023, REQ-B-004 to 006, REQ-R-003 to 006): Not implemented in Phase 1; SA must design for extensibility.

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9. Open Questions (Blocking for SAD)

#	Question	Priority	Blocks	Owner to answer
Q1	**Platform** — iOS native / PWA / React Native / Flutter	P0 (blocking)	All UI code + deploy	SA + Owner
Q2	**Backend** — Firebase / Supabase / custom	P0 (blocking)	Persistence + Phase 2 auth	SA + Owner
Q3	**Data model finalization** — attribute-level for User/Submission/Plan/Day/Item	P0 (blocking)	Backend + DDD	SA
Q4	Notification method — email only / add push	P1	E8 interface	SA + Owner
Q5	Analytics platform	P1	E6 interface	SA
Q6	Deployment infra + region	P1	Deploy Arch	SA
Q7	Testing framework choice	P1	Test Strategy	SA
Q8	Feature flag mechanism	P2	Phase 2 rollout	SA
Q9	LLM API choice (Phase 2)	P3	Phase 2 arch	SA
Q10	Payment gateway (Phase 2)	P3	Phase 2 arch	SA + Owner
Q11	Legal review requirement	P3	Phase 2 launch	Owner + legal

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10. Decision Log

Date	Decision	Rationale	Made by
2026-07-25	SRS v0.1 draft created from REQ + HND	Initial SA phase output	SA (via /systems-analyst)
2026-07-25	Data model designed as polymorphic Item per domain	Extensibility for Phase 2 + flex per user	SA (draft)
2026-07-25	Analytics event catalog specified (§7.4)	Ensures no PII leaks + consistent naming	SA (draft)
2026-07-25	Q1/Q2 marked as blocking; SAD ต้องรอ	Cannot commit tech stack without decision	SA

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11. Change Control

SRS changes require: 1. Documented change reason (new REQ / discovered constraint / SA analysis) 2. Impact on downstream: SAD, DDD, API, DB, code 3. Review by PO/PM 4. Version bump + regenerate PDF/DOCX 5. Update traceability matrix if scope changed

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12. Next Steps for SA Phase

1. **Answer Q1/Q2** (with owner) → ADR entries
2. **Write SAD** using `assets/sad-template.md` — pick pattern, run quality attribute analysis
3. **Write DDD** — flesh out §6.1 logical model → attributes + business rules
4. **Write API Spec** — endpoints from §4 features + §3.3 interfaces
5. **Write DB Schema** — persistence model from DDD
6. **Write Security Architecture** — auth (Phase 1 admin, Phase 2 user), encryption, PDPA compliance flow
7. **Write Deployment Architecture** — based on Q2 backend decision
8. **Write Test Strategy** — coverage plan per §5, §7.4 events